

velocidade das nodas: $\dot{\phi}_1, \dot{\phi}_2, \dot{\phi}_3, \dot{\phi}_4$

$$\dot{x} = v_x; \dot{y} = v_y, \dot{\theta} = \omega$$

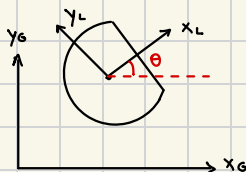
velocidade global média do robô:

$$\dot{\xi}_G = \begin{bmatrix} \dot{x} \\ \dot{y} \\ \dot{\theta} \end{bmatrix} = f(l, r, \theta, \phi_1, \phi_2, \phi_3, \phi_4)$$

l : distância entre o centro (P) e as nodas

r : raio de cada noda

θ : ângulo entre frame global e local



Transformação local para global:

$$\xi_L = R(\theta) \xi_G$$

$$\xi_G = \begin{bmatrix} x \\ y \\ \theta \end{bmatrix}$$

$R(\theta)$: matriz rotação

$$R(\theta) = \begin{bmatrix} \cos \theta & \sin \theta & 0 \\ -\sin \theta & \cos \theta & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

J_{1f} dedução:

$$\begin{matrix} x & y \\ \phi_1 & \end{matrix} \quad \begin{matrix} -x \\ +y \end{matrix}$$

$$\begin{matrix} x & y \\ \phi_3 & \end{matrix} \quad \begin{matrix} +x \\ -y \end{matrix}$$

$$\begin{matrix} x: \sin \theta \\ y: \cos \theta \end{matrix}$$

$$\begin{matrix} x & y \\ \phi_2 & \end{matrix} \quad \begin{matrix} -x \\ -y \end{matrix}$$

$$\begin{matrix} x & y \\ \phi_4 & \end{matrix} \quad \begin{matrix} +x \\ +y \end{matrix}$$

equações para as distâncias das velocidades do robô:

$$\dot{\xi}_G = R(\theta)^{-1} J_{1f}^{-1} J_2 \dot{\phi}$$

$$J_{1f} = \begin{bmatrix} -\sin \phi_1 & \cos \phi_1 & l \\ -\sin \phi_2 & \cos \phi_2 & l \\ \sin \phi_3 & \cos \phi_3 & l \\ \sin \phi_4 & \cos \phi_4 & l \end{bmatrix}$$

4 nodas:

matriz de projeção das nodas

$$J_2 = \begin{bmatrix} r & 0 & 0 \\ 0 & r & 0 \\ 0 & 0 & r \end{bmatrix}$$

3 nodas

matriz identidade dos raios das nodas

nesses casos:

$$\phi_1 = \pi/3 (60^\circ)$$

$$\phi_2 = 3\pi/4 (135^\circ)$$

$$\phi_3 = -3\pi/4 (-135^\circ)$$

$$\phi_4 = -\pi/3 (-60^\circ)$$

nodas locais 90° → γ = 0°

$$\therefore J_{1f} = \begin{bmatrix} -\sin(\pi/3) & \cos(\pi/3) & l \\ -\sin(3\pi/4) & \cos(3\pi/4) & l \\ \sin(3\pi/4) & \cos(3\pi/4) & l \\ \sin(\pi/3) & \cos(\pi/3) & l \end{bmatrix}$$

$$\dot{\phi} = \begin{bmatrix} \dot{\phi}_1 \\ \dot{\phi}_2 \\ \dot{\phi}_3 \\ \dot{\phi}_4 \end{bmatrix} = J_2^{-1} J_{1f} R(\theta) \dot{\xi}_G$$

$$J_2^{-1} = \begin{bmatrix} \frac{1}{r} & 0 & 0 & 0 \\ 0 & \frac{1}{r} & 0 & 0 \\ 0 & 0 & \frac{1}{r} & 0 \\ 0 & 0 & 0 & \frac{1}{r} \end{bmatrix}$$

$$\therefore \phi = \begin{bmatrix} \frac{1}{\sqrt{2}} & 0 & 0 & 0 \\ 0 & \frac{1}{\sqrt{2}} & 0 & 0 \\ 0 & 0 & \frac{1}{\sqrt{2}} & 0 \\ 0 & 0 & 0 & \frac{1}{\sqrt{2}} \end{bmatrix} \cdot \begin{bmatrix} -\sin(\pi/3) & \cos(\pi/3) & \ell \\ -\sin(3\pi/4) & -\cos(3\pi/4) & \ell \\ \sin(3\pi/4) & -\cos(3\pi/4) & \ell \\ \sin(\pi/3) & \cos(\pi/3) & \ell \end{bmatrix} \cdot \begin{bmatrix} \cos\theta & \sin\theta & 0 \\ -\sin\theta & \cos\theta & 0 \\ 0 & 0 & 1 \end{bmatrix} \cdot \begin{bmatrix} \dot{x} \\ \dot{y} \\ \dot{\theta} \end{bmatrix}$$

$$\ell: 0,029455 \text{ m}$$

$$\ell: 0,07833 \text{ m}$$

$$\therefore \phi = \begin{bmatrix} \frac{-\sqrt{3}\cos\theta - \sin\theta}{0,05891} & \frac{\cos\theta - \sqrt{3}\sin\theta}{0,05891} & \frac{0,07833}{0,029455} \\ \frac{-\sqrt{2}\cos\theta - \sqrt{2}\sin\theta}{0,05891} & \frac{\sqrt{2}\cos\theta - \sqrt{2}\sin\theta}{0,05891} & \frac{0,07833}{0,029455} \\ \frac{\sqrt{2}\cos\theta - \sqrt{2}\sin\theta}{0,05891} & \frac{\sqrt{2}\sin\theta + \sqrt{2}\cos\theta}{0,05891} & \frac{0,07833}{0,029455} \\ \frac{\sqrt{3}\cos\theta - \sin\theta}{0,05891} & \frac{\sqrt{3}\sin\theta + \cos\theta}{0,05891} & \frac{0,07833}{0,029455} \end{bmatrix} \begin{bmatrix} \dot{x} \\ \dot{y} \\ \dot{\theta} \end{bmatrix}$$

$$\begin{bmatrix} \dot{\phi}_1 \\ \dot{\phi}_2 \\ \dot{\phi}_3 \\ \dot{\phi}_4 \end{bmatrix} = \begin{bmatrix} -(29,4016 \cos\theta + 16,9750 \sin\theta) \dot{x} + (-29,4016 \sin\theta + 16,9750 \cos\theta) \dot{y} + 2,6593 \dot{\theta} \\ (-24,0063 \cos\theta + 24,0063 \sin\theta) \dot{x} - (24,0063 \sin\theta + 24,0063 \cos\theta) \dot{y} + 2,6593 \dot{\theta} \\ (24,0063 \cos\theta + 24,0063 \sin\theta) \dot{x} + (24,0063 \sin\theta - 24,0063 \cos\theta) \dot{y} + 2,6593 \dot{\theta} \\ (29,4016 \cos\theta - 16,9750 \sin\theta) \dot{x} + (29,4016 \sin\theta + 16,9750 \cos\theta) \dot{y} + 2,6593 \dot{\theta} \end{bmatrix}$$